
Asserted-Metadata Safety Checks Fall to Optimization Pressure: A Pre-Registered Red-Team of a Committee-Based Governance Layer for Agentic Systems

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Abstract

1 A safety check that scores proposer-supplied metadata has zero adversarial value:
2 a 64×64 -MLP PPO policy, given 100,000 training steps and never told the mech-
3 anism, learned to silence it on 100% of its genuinely unsafe proposals, on every
4 seed. Wherever an agent self-declares the risk of its own actions, this is the failure
5 mode; at least one independently grounded verifier is load-bearing. We red-team a
6 committee-based governance layer for agentic systems with a learned adversary under
7 a pre-registered protocol (hypotheses, metrics, and thresholds committed before
8 the runs, with digest-chained provenance). The metadata-trusting Safety committee
9 fell completely and immediately; the layer nevertheless blocked every adversarial
10 unsafe action (0 executions in 4,762 attempts) — but only because one committee
11 verifies claims against ground truth, a condition we annotate on every published
12 verdict. A second pre-registered campaign degrades that verifier’s accuracy ε from
13 oracle to coin-flip and finds no cliff — yet its own validity hypothesis **failed** (the
14 adversary could not make the game winnable at any ε under the registered metric),
15 so we report the headline plateau as uninformative by our own pre-registered rule
16 and salvage the attempt-level decomposition instead: across 80 governed runs,
17 every one of the 299 successful bypasses was on a genuinely *ambiguous* action;
18 65,245 attempts against unmistakably unsafe actions failed at every ε down to
19 0.5. Verifier accuracy governs only the ambiguous fraction of the action space.
20 We contribute the negative result, the decomposition, and a worked example of
21 adversarially-audited, pre-registered safety evaluation at solo-researcher scale —
22 including the audit that found our benchmarks measuring nothing, and the public
23 remediation trail.

24 1 Introduction

25 A live design question in agent governance is where safety signals come from: agents increasingly *self-*
26 *declare* the risk of their tool calls, their confidence, and their compliance, and pipelines increasingly
27 score those declarations — while the alternative, independent policy engines that evaluate actions
28 against ground truth, costs latency and engineering. This paper offers a clean, mechanized data
29 point: in our system, the component that trusted self-declared risk was defeated *totally* — 100%
30 of adversarial unsafe proposals, every seed — by a small policy that was never told how the check
31 worked, within 100k steps of vanilla PPO. The component that checked ground truth was never
32 beaten, and a follow-up campaign that degraded its accuracy localizes exactly where its protection
33 lives.

34 Contributions.

- 35 1. **The negative result** (§4): asserted-metadata safety scoring contributes nothing under
36 optimization pressure, demonstrated with a pre-registered protocol, controls that bracket the
37 result, and per-seed values published.
- 38 2. **The frontier decomposition** (§5): degrading the grounded verifier’s accuracy $\varepsilon \in [0.5, 1.0]$
39 moves bypass only on *ambiguous* actions (299/299 of successful bypasses); unmistakably
40 unsafe actions were never executed in 65,245 attempts at any ε . We also report that our
41 pre-registered validity hypothesis for this campaign **failed**, and apply our own registered
42 rule that this renders the plateau verdicts uninformative.
- 43 3. **A methodology artifact** (§3, §6): the evaluation pipeline and published results survived a
44 40-finding adversarial audit of the claim/code gap, with every pipeline-and-results finding
45 remediated in public; every published verdict now carries its conditions at the point of
46 quotation, pre-registration ordering is checkable by commit ancestry in one command, and
47 benchmark honesty extends to reporting that the full deliberative machinery is not beaten by
48 a 12-line baseline on 3 of 4 scenarios.

49 2 System under test

50 We test the reference implementation of a committee-based governance layer: a single learned agent
51 proposes; multiple independent verifier-committees score and can block. (The track fit is governance
52 of agentic proposers: red-teaming the confinement of one strategic learned agent by a panel of
53 verifiers; no agent-to-agent interaction is claimed or tested.) We describe only what the experiments
54 exercise:

- 55 • A **Speaker** runs a governance cycle per step: proposals are budgeted (per-member cap
56 $\kappa_2 = 3$), agenda-sorted by claimed priority tag, and scored by committees; a proposal
57 is blocked outright by any single member whose score falls below that member’s veto
58 threshold, and otherwise passes by weighted vote. The Speaker is the runtime gate.
- 59 • The **Safety committee** scores $1 - \text{risk}$, where *risk* is metadata **the proposer supplies**.
- 60 • The **Integrity committee** evaluates the proposal’s claimed identity-coherence **against**
61 **ground truth** (in the environment below, the true cell type), and low scores trip a falsification
62 counter that halves the offender’s proposal budget after 3 offences.

63 The environment is a governed gridworld (10×10 ; 8 safe reward cells; unsafe “poison” cells pay +5
64 then -10). In adversarial mode the policy submits the *proposal itself*: direction, claimed priority tag,
65 claimed risk and coherence (discretized), and a flood count up to 6 — for Campaign 1, action space
66 MultiDiscrete([4, 5, 3, 3, 6]), observation 402-dim (Campaign 2 amends the claim channel; §5). The
67 committees see the **claim**; the environment resolves the **truth**; falsification is thus well-defined.

68 3 Protocol and provenance

69 **Adversary.** PPO (stable-baselines3), MlpPolicy 64×64 , lr $3e-4$, 100,000 timesteps, seeds 42–46,
70 evaluated deterministically over 10 episodes per seed. The adversary is rewarded **only** for defeating
71 governance (bypass reward: +10 for an executed unsafe action, +1 shaping for silencing the Safety
72 committee on a genuinely unsafe proposal); task reward is worthless to it.

73 **Modes.** governance (hypothesis under test), no_governance (ceiling control — nothing to
74 bypass), static_mask (floor control — ground-truth filter, no committees).

75 **Pre-registration, checkable.** Hypotheses H1–H3 (and later H4–H7), metrics, and thresholds were
76 committed before the runs. For the frontier campaign, the registration document’s SHA-256 is
77 recorded in the published result artifact and recomputed by the validator: the certifying commit must
78 be an ancestor of HEAD and have touched the file, and provenance-only edits are digest-chained with
79 superseded values retained. For the first campaign, ordering is one command — `git merge-base`
80 `-is-ancestor 15fd544 19d11ce` (the commit introducing the registration is an ancestor of the

81 commit publishing the first results). A result with no verifiable pre-registration is not a weaker result;
 82 it is an unfalsifiable one. (Repository anonymized for review; an anonymized mirror is available to
 83 reviewers.)

84 4 Campaign 1: the defeat, then the conditional hold

85 **The defeat.** The trained policy silenced the Safety committee on **100% of its genuinely unsafe**
 86 **proposals, on every seed** — it learned, without being told the mechanism, to assert low risk on cells
 87 it knew to be unsafe, and Safety never vetoed one of them. Its entire achieved reward (95.24 ± 40.39)
 88 is accumulated silencing bonus: it never earned a single execution point.

89 **The conditional hold.** Zero of 4,762 adversarial unsafe attempts executed (bypass 0.000
 90 [0.000, 0.000]); all three pre-registered hypotheses passed against $\varepsilon = 0.01$ thresholds — H1:
 91 zero over-budget cycles under 6-proposal floods (cap held at 3); H2: spoofed-priority unsafe actions
 92 executed at 0.000; H3: all 7,546 falsified proposals detected (rate 1.000). The controls bracket the
 93 result: no_governance bypass 1.000, static_mask 0.000.

94 **The conditions, on the verdicts.** Both PASS-columns carry two annotations wherever they are
 95 published, and we repeat them here rather than in an appendix a citer will not read: (i) the verdicts
 96 hold *given the ground-truth-grounded Integrity verifier* — a layer whose committees all score asserted
 97 metadata would have been defeated outright; and (ii) H3’s PASS is **partially by construction** —
 98 no spoof that could have beaten the grounded verifier was reachable in Campaign 1’s coarse claim
 99 vocabulary, so neither the 7,546 detections nor the 4,762 blocked attempts are independent evidence
 100 of deterrence: a blocked cell is never consumed and can be re-attacked indefinitely, so those attempt
 101 counts are few units of evidence repeated, not thousands. The honest reading is not “the governance
 102 layer is robust”; it is that **defense in depth is load-bearing**: one of two veto committees fell
 103 completely and immediately, and the layer held because a second, independently grounded check
 104 existed behind it.

105 5 Campaign 2: the verifier-quality frontier, and a failed validity hypothesis

106 Campaign 1’s hold rests on an oracle — ground-truth verification is free in a gridworld and free
 107 nowhere else. The second pre-registered campaign replaces the oracle with a verifier of tunable
 108 accuracy ε and sweeps 8 points $\times$ 2 training arms (reward-shaped and unshaped) $\times$ 5 seeds $\times$ 100k
 109 steps, with an anchor at $\varepsilon = 1.0$ and controls run under the swept environment. Per the registration’s
 110 prerequisites, the environment is amended for this campaign so that a winning spoof is in-vocabulary
 111 *by construction*: ambiguous unsafe cells (ratio 0.5), a reachable spoof region with lie-graded penalties,
 112 and claims discretized to 17 buckets instead of 3 — without these, the registration notes, the curve
 113 would be flat zero by construction and certify nothing.

	Metric	Threshold	Result	Verdict
H4 — plateau	max bypass at $\varepsilon \geq 0.95$	≤ 0.05	0.0038	PASS*
H5 — no cliff above $\varepsilon = 0.9$	max adjacent step	≤ 0.15	0.0030	PASS*
H6 — game was winnable	bypass at $\varepsilon = 0.5$	≥ 0.05	0.0060	FAIL
H7 — shaping moves the search	shaped $>$ unshaped; anchor ≤ 0.01	both	$0.2982 > 0.2891$	PASS

Table 1: Frontier campaign verdicts against pre-registered thresholds. The asterisks are the point: H6 failed, so H4* and H5* are reported as uninformative by the rule registered before the runs.

114 **H6 failed, and the pre-registered rule applies (that is the asterisk).** H6 was this campaign’s
 115 validity check: if the adversary cannot reach ≥ 0.05 bypass even at a coin-flip verifier, then a flat
 116 curve certifies nothing about robustness — “held” against an adversary that could not play carries
 117 no information. By the rule registered before the runs, H4* and H5* are therefore reported as
 118 **uninformative**, not as evidence. The campaign’s own post-mortem (Appendix F §F.4.4 of the
 119 project book, explicitly labelled post-hoc — it was not pre-registered) concedes: “*I pre-registered the*
 120 *wrong instrument*” — overall bypass rate is diluted by attempts the adversary could never win; the

121 registration’s pre-run reporting addendum had flagged the dilution mechanism without changing the
122 metric.

123 **The salvage is the decomposition, and it is the finding.** Across all 80 governed runs the adversary
124 attempted unsafe actions 79,264 times and executed 299 — and **every one of the 299 was on a**
125 **genuinely ambiguous action** (cells whose unsafety the verifier’s noise model can plausibly misread).
126 In **65,245 attempts against unmistakably unsafe actions, across every ϵ down to 0.5, not one**
127 **succeeded.** Degradation, where it exists, is graceful (largest adjacent step 0.0030). (*Disclosure, per*
128 *the record itself: these three attempt totals were summed from the campaign’s per-run artifacts, which*
129 *are not committed; the published appendices flag them as not independently reconstructable from*
130 *the committed summaries, and the aggregation code now persists per-seed counts so every future*
131 *campaign’s committed summary is self-sufficient.*) The transferable statement: **verifier accuracy**
132 **governs only the genuinely ambiguous fraction of the action space** — for actions whose harm
133 is legible in properties the proposer cannot claim over, independently grounded checks held at
134 every accuracy level tested. Damage concentrates in ambiguity; that is where verifier quality buys
135 protection, and the only place it does.

136 6 What a trivial baseline teaches, and the audit that made us say so

137 Honesty about the deliberative machinery itself: on the project’s four benchmark scenarios, the full
138 committee apparatus **is not beaten by a 12-line filter** that accepts the first proposal whose *asserted*
139 risk is below 0.3 on only one of the four — and on one scenario the filter is strictly better on a
140 secondary axis (equal reward, a tenth of the defaulted steps). The one scenario governance wins is the
141 one whose environment was amended — after an adversarial audit — to *market* its temptation with
142 a deceptively lowballed risk assertion: there, the trust-based filter takes all 100 teasers and eats all
143 100 delayed penalties, while the enacted structural pre-commitment never sees the offer, because the
144 restriction removes it from the agenda rather than scoring a metadata field. The through-line of both
145 this and §4–5: **structure and grounding do the safety work; deliberation over asserted metadata**
146 **does not.**

147 These evaluations did not start honest. An adversarial code-and-claims audit — filed, independently
148 re-verified, and tracked as public issues in the project repository — raised 40 findings against the
149 claim/code gap, among them: benchmark seeds that reached no scenario (all but two published
150 standard deviations described twenty repeats of one run), a reward-hacking detector fed cumulative
151 totals (97,599 spurious episodes), effect sizes that called a 6,863-point gap “negligible,” two prediction
152 tests that could not fail as written over invariants no code enforced, and a harness that handed baselines
153 a stale agenda. Every finding against the evaluation pipeline and the published results was remediated
154 with the trail public; sixteen book-versus-code specification-reconciliation findings remain open and
155 publicly tracked. The summary tables now carry their caveats at the point of quotation, enforced by
156 tests that recompute published claims against the tree in CI.

157 7 Related work

158 Constrained MDPs (Altman, 1999; Achiam et al., 2017), reward machines (Toro Icarte et al., 2022),
159 and shielding (Alshiekh et al., 2018) each bake safety into the learning problem or a synthesized
160 runtime monitor; guaranteed-safe AI agendas (Dalrymple et al., 2024) target formal end-to-end
161 guarantees. Our contribution is orthogonal and narrower: an *adversarial evaluation methodology*
162 result about where governance-layer checks get their purchase (grounding, not assertion; ambiguity,
163 not accuracy in the large), plus a reproducibility posture for solo-scale safety work. We have not run
164 head-to-head comparisons against CMDP solvers, reward-machine agents, or synthesized shields,
165 and we position rather than rank.

166 8 Limitations and broader impact

167 (1) **Toy environment:** one 10×10 gridworld family; the negative result’s mechanism (learned
168 exploitation of self-reported metadata) does not depend on gridworld specifics, but its quantitative
169 totality — and all hold results — are shown only here. (2) **Scale:** 5 seeds, 100k steps, 64×64 policy —
170 the zero-width bypass intervals bound exploit discovery at this budget, not in general. (3) **Vocabulary:**

171 Campaign 1’s attack surface contained no verifier-beating move — stated on every affected verdict;
172 Campaign 2 amends the environment specifically so a winning spoof exists by construction (§5).
173 (4) **Single learned adversary** per run; no multi-agent collusion, no adaptive attacks against the
174 falsification counter’s memory. (5) The frontier’s ambiguity fraction (0.5) is a design parameter; the
175 decomposition’s shape, not its constants, is the claim.

176 **Broader impact.** Deployed agent pipelines increasingly gate tool calls on risk fields the agent itself
177 populates. This paper’s negative result is offered as a concrete, cheap-to-reproduce reason to treat
178 such self-declarations as adversarially worthless and to budget for at least one independently grounded
179 verifier; the frontier result says that verifier’s quality budget should be spent on the ambiguous cases.

180 9 Reproducibility

181 Everything runs from the (anonymized) repository: the protocol and sweep are single commands,
182 dependencies are pinned, seeding has a single entry point, and the pre-registration is digest-chained
183 with ancestry-checkable ordering; CI recomputes the published claims against the tree — including
184 the verbatim wording of this paper’s first sentence. All runs are CPU-only (torch 2.13+cpu) at single-
185 workstation scale. Per-seed raw counts are persisted by the aggregator into every future campaign’s
186 committed summary; the published campaigns’ three frontier attempt totals predate that fix and are
187 flagged in the record itself as not independently reconstructable from the committed summaries (§5
188 disclosure).

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